9 State Space Analysis

INTRODUCTION:

To study the behaviour of an LTI control system, we use the conventional system in which the transfer function is the ratio of the output to input with initial conditions zero. It has limitations which are listed below-

(i) We cannot obtain precise solution of a system in which initial condition plays an important role.

(ii) For MIMO (multiple inputs multiple outputs) system, it is not a convenient approach.

(iii) For higher order system, the conventional process becomes troublesome.

(iv) It is applicable to LTI systems only.

(v) It gives result for some specific types of input like step, ramp etc.

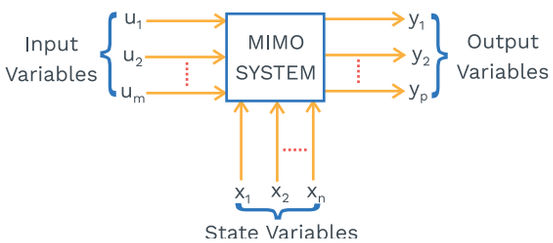
(vi) Controllability and observability can not be determined.

So, the state variable analysis overcomes all the limitations listed above and it can also be termed as modern method.

State Model of Linear System

Let us take a MIMO (multiple input multiple output) system of nth order where-

m = Number of inputs
p = Number of outputs
n = Number of state variables



Block diagram representation of a general control system

Definitions

(i) State: Minimal set of variables which completely determine the behaviour of system at any time $t \geq t_0$.

(ii) State Variables: The variables involved to determine the state of a dynamic system x(t). It can be represented as $x_1(t)$, $x_2(t)$,... etc.

(iii) State Vector: For describing the behaviour of the system, we consider 'n' state variables. These 'n' variables are considered as 'n' components of a vector x(t) which is called state vector at time 't'.

(iv) State Space: The space whose co-ordinate axis are nothing but the 'n' state variables that varies with time.

Thus, Input vector, $u(t) = \begin{bmatrix} u_1(t) \\ u_2(t) \\ \vdots \\ u_m(t) \end{bmatrix}_{m\times 1}$

Output vector, $y(t) = \begin{bmatrix} y_1(t) \\ y_2(t) \\ \vdots \\ y_p(t) \end{bmatrix}_{p\times 1}$

State vector, $x(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix}_{n\times 1}$

Rack your Brain

State space analysis is applicable even if initial conditions are ____.

State Equation:

$$\dot{x}(t) = Ax(t) + Bu(t)$$

Where x(t) = State vector of n × 1 order
u(t) = Input vector of m × 1 order
A = System matrix of n × n order
B = Input matrix of n × m order
In matrix form-

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_n \end{bmatrix}_{n\times 1} = [A]_{n\times n} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}_{n\times 1} + [B]_{n\times m} \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_m \end{bmatrix}_{m\times 1}$$

Output Equation:

$$y(t) = Cx(t) + Du(t)$$

Where, y(t) = Output vector of p × 1 order
C = Output matrix or observation matrix of p × n order
D = Direct transmission matrix of p × m order

In matrix form

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_p \end{bmatrix}_{p\times 1} = [C]_{p\times n} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}_{n\times 1} + [D]_{p\times m} \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_m \end{bmatrix}_{m\times 1}$$

Rack your Brain

What is the order of system matrix?

State Space Representation Using Phase Variables

(i) State model from differential equation-

$$y^n(t) + a_{n-1}y^{n-1}(t) + ...a_1\dot{y}(t) + a_0y(t) = b_0u(t) + b_1\dot{u}(t) + ... + b_mu^m(t)$$

where, y(t) = Output variable

u(t) = Input variable

$$x_1(t) = y(t), x_2(t) = \dot{y}(t) = \dot{x}_1(t)$$

$$x_3(t) = \ddot{y}(t) = \ddot{x}_1(t) = \dot{x}_2(t)...$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ : \\ \dot{x}_n \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 & 0.....0 \\ 0 & 0 & 1.....0 \\ -a_0 & -a_1..... & -a_{n-1} \end{bmatrix}}_{\text{Bush form or companion form}} \begin{bmatrix} x_1 \\ x_2 \\ : \\ x_n \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ : \\ b_0 \end{bmatrix} u(t)$$

Example-1.

Construct the state model from the differential equation given below.

$$\frac{d^3y(t)}{dt^3} + \frac{6d^2y(t)}{dt^2} + \frac{8dy(t)}{dt} + 2y(t) = 10u(t)$$

Solution:

$$x_1(t) = y(t)$$

$$x_2(t) = \dot{x}_1(t) = \dot{y}(t) = \frac{dy(t)}{dt}$$

$$x_3(t) = \dot{x}_2(t) = \ddot{y}(t) = \frac{d^2y(t)}{dt^2}$$

$$\dot{x}_3(t) = -2x_1(t) - 8x_2(t) - 6x_3(t) + 10u(t)$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & -8 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 10 \end{bmatrix} u(t)$$

$$y(t) = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} x(t)$$

(ii) State Model From Transfer Function

(a) Signal Flow Graph-

Let us consider a system transfer function given by-

$$T(s)=\frac{b_0+sb_1+s^2b_2+s^3b_3}{a_0+sa_1+s^2a_2+s^3}=\frac{Y(s)}{U(s)}$$

Divide both numerator and denominator by highest power of s-

$$T(s)=\frac{\frac{b_0}{s^3}+\frac{b_1}{s^2}+\frac{b_2}{s}+b_3}{1+\frac{a_2}{s}+\frac{a_1}{s^2}+\frac{a_0}{s^3}}$$

$$=\frac{b_3+\frac{b_2}{s}+\frac{b_1}{s^2}+\frac{b_0}{s^3}}{1-\left[-\frac{a_2}{s}-\frac{a_1}{s^2}-\frac{a_0}{s^3}\right]}$$

Mason gain formula

$$=\frac{T_1\Delta_1+T_2\Delta_2+T_3\Delta_3+T_4\Delta_4}{\left(1-\left[\sum \text{All loop gains}\right]+\left[\sum \text{gain products of 2 non-touching loops}\right]\right)}$$

If loops touch the forward path then-

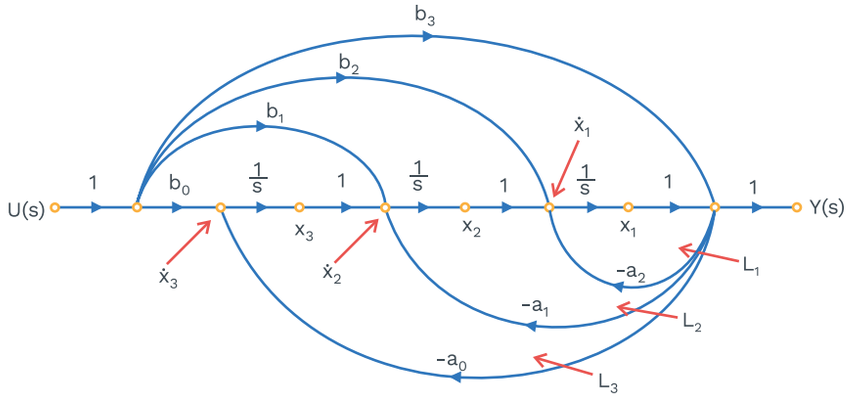
$\Delta_1=\Delta_2=\Delta_3=\Delta_4=1$

The loop gains are-

$$L_1=-\frac{a_2}{s},L_2=-\frac{a_1}{s^2},\quad L_3=-\frac{a_0}{s^3}$$

Therefore, the forward path gains are-

$$T_1=b_3,\ T_2=\frac{b_2}{s},T_3=\frac{b_1}{s^2},\ T_4=\frac{b_0}{s^3}$$



$$\dot{x}_1=b_2u+x_2-a_2y$$

$$\dot{x}_2 = b_1u + x_3 - a_1y$$

$$\dot{x}_3 = b_0u - a_0y$$

$$y = b_3u + x_1$$

$$\dot{x}_1 = -a_2\,x_1 + x_2 + \left[b_2 - a_2b_3\right]u(t)$$

$$\dot{x}_2 = -a_1x_1 + x_3 + \left[b_1 - a_1b_3\right]u(t)$$

$$\dot{x}_3 = -a_0x_1 + \left[b_0 - a_0b_3\right]u(t)$$

$$\dot{x}(t) = A\,x(t) + Bu(t)$$

$$A = \begin{bmatrix} -a_2 & 1 & 0 \\ -a_1 & 0 & 1 \\ -a_0 & 0 & 0 \end{bmatrix}, B = \begin{bmatrix} b_2 - a_2b_3 \\ b_1 - a_1b_3 \\ b_0 - a_0b_3 \end{bmatrix}$$

$$y(t) = Cx(t) + Du(t)$$

$$C = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}, D = b_3$$

Example-2.

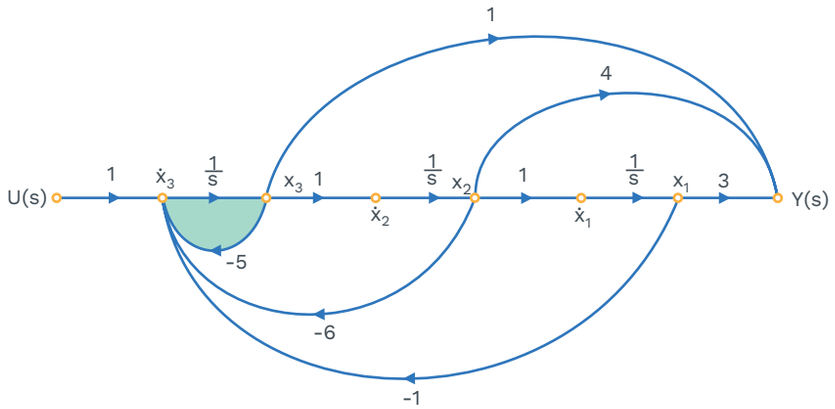
A feedback system is characterized by closed loop transfer function given by-

$$T(s) = \frac{s^2 + 4s + 3}{s^3 + 5s^2 + 6s + 1}$$

Draw SFG and obtain state model.

Solution:

$$T(s) = \frac{\frac{1}{s} + \frac{4}{s^2} + \frac{3}{s^3}}{1 - \left[-\frac{5}{s} - \frac{6}{s^2} - \frac{1}{s^3}\right]}$$



$$\dot{x}_1 = x_2, \dot{x}_2 = x_3$$

$$\dot{x}_3 = -x_1 - 6x_2 - 5x_3 + u$$

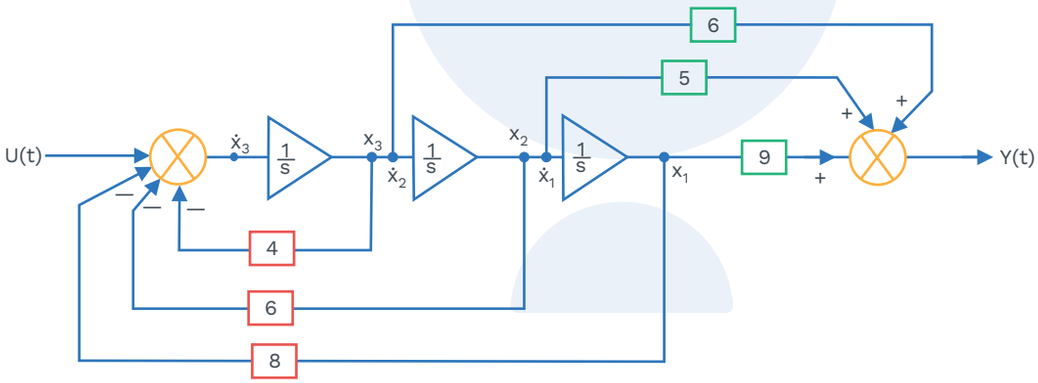
$$y = 3x_1 + 4x_2 + x_3$$

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -6 & -5 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, C = \begin{bmatrix} 3 & 4 & 1 \end{bmatrix}, D = 0$$

(b) Direct Decomposition form-

$$\frac{Y(s)}{U(s)} = \frac{6s^2 + 5s + 9}{s^3 + 4s^2 + 6s + 8}$$

$$s^3 + 4s^2 + 6s + 8 = \{([s + 4]s + 6)s + 8\}$$



$$\dot{x}_3 = u(t) - 8x_1(t) - 6x_2(t) - 4x_3(t)$$

$$y(t) = 9x_1(t) + 5x_2(t) + 6x_3(t)$$

$$\dot{x}(t) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -8 & -6 & -4 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u(t)$$

$$y(t) = \begin{bmatrix} 9 & 5 & 6 \end{bmatrix} x(t)$$

Canonical Form

It is also known as Foster form or normal form. This method is based on the partial fraction expansion.

$$T(s) = \frac{b_0 s^m + b_1 s^{m-1} + b_2 s^{m-2} + ... + b_m}{(s + a_1)(s + a_2)....(s + a_n)}$$

Case:1

If the degree 'm' is less than 'n' (m < n), then T(s) can be expressed in partial fraction expansion as,

$$T(s) = \frac{c_1}{s+a_1} + \frac{c_2}{s+a_2} + ... + \frac{c_n}{s+a_n} = \sum_{i=1}^{n} \frac{c_i}{s+a_i}$$

$$\dot{x}_1 = -a_1 x_1 + u$$

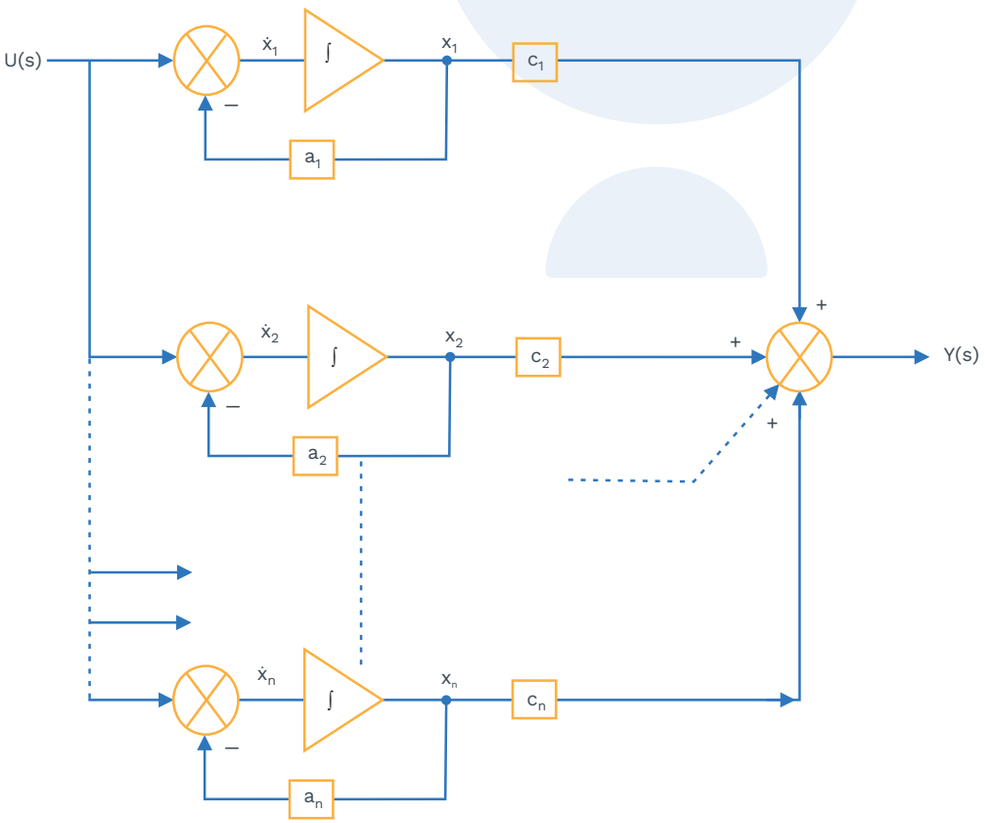
$$\dot{x}_2 = -a_2 x_2 + u$$

$$\vdots$$

$$\dot{x}_n = -a_n x_n + u$$

$$y = c_1 x_1 + c_2 x_2 + ...c_n x_n$$

$$\dot{x} = Ax + Bu \text{ and } y = Cx + Du$$



Foster form block diagram

$$A = \begin{bmatrix} -a_1 & 0 & 0\ldots & 0 \\ 0 & -a_2 & 0\ldots & 0 \\ \vdots & & & \\ 0 & 0 & 0\ldots & -a_n \end{bmatrix} \quad B = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$$

$$C = \begin{bmatrix} c_1 & c_2 & \ldots & c_n \end{bmatrix} \text{ and } D = 0$$

Gray Matter Alert!!!

This method is valid for non-repeated real roots.
Matrix A is diagonal.
The diagonal elements are poles of transfer function.

Case:2

If the degree m = n

$$T(s) = \frac{N(s)}{D(s)} = c_0 + \sum_{i=1}^{n} \frac{c_i}{s + a_i}$$

c_0 = Constant obtained by dividing N(s) by D(s).

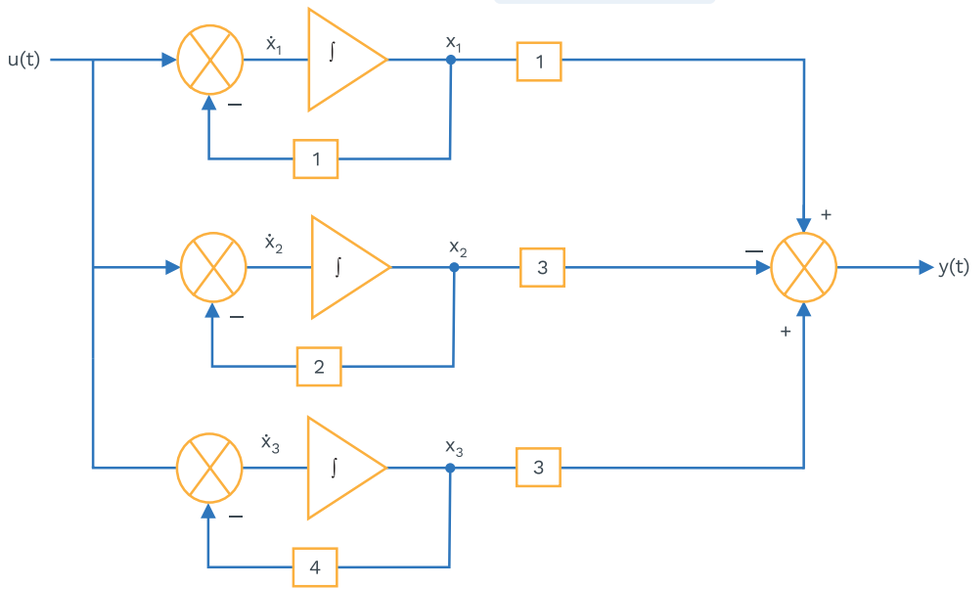
Example-3.

Obtain the state model by canonical form of a system whose transfer function is-

$$\frac{s^2 + 2}{(s+1)(s+2)(s+4)}$$

Solution:

$$\frac{s^2 + 2}{(s+1)(s+2)(s+4)} = \frac{1}{s+1} - \frac{3}{s+2} + \frac{3}{s+4}$$



$$\dot{x}_1 = u(t) - x_1(t), \dot{x}_2 = u(t) - 2x_2(t)$$

$$\dot{x}_3 = u(t) - 4x_3(t)$$

$$y(t) = 1x_1(t) - 3x_2(t) + 3x_3(t)$$

State model-

$$\dot{x} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -4 \end{bmatrix} x + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & -3 & 3 \end{bmatrix} x + \begin{bmatrix} 0 \end{bmatrix} u$$

Jordan's Canonical Form

When the roots are repeated then matrix A is said to be in Jordan's canonical form.

Example-4.

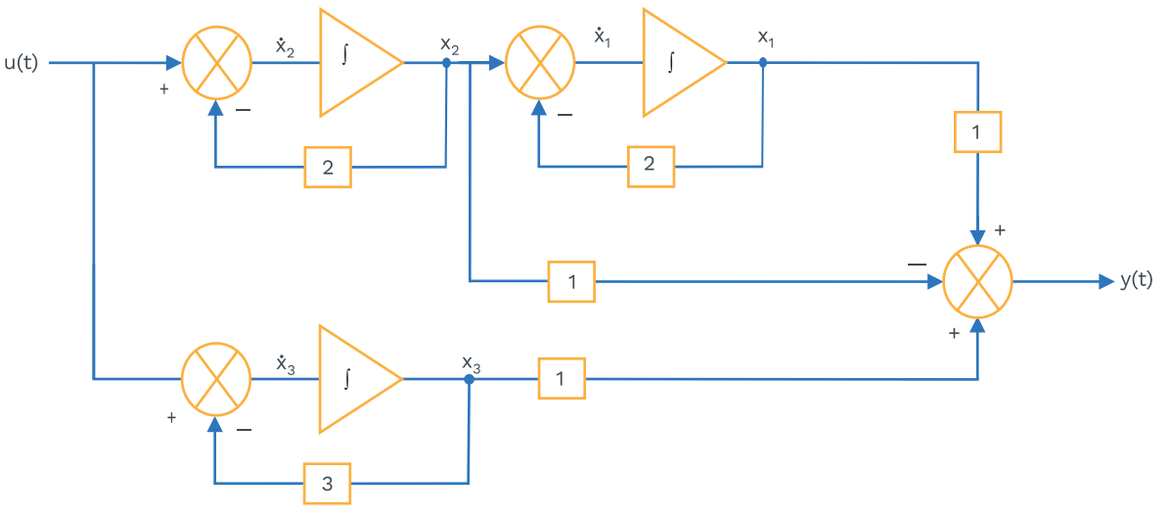
Obtain the state model in Jordan's canonical form of a system whose transfer function is-

$$T(s) = \frac{1}{(s+2)^2(s+3)}$$

Solution:

$$T(s) = \frac{A}{(s+2)^2} + \frac{B}{s+2} + \frac{C}{s+3}$$

$$= \frac{1}{(s+2)^2} - \frac{1}{s+2} + \frac{1}{s+3}$$



Gray Matter Alert!!!

In Jordan form,

(1) Principle diagonal consists of all the poles of transfer function with repeated poles 'r' times.

(2) r-1 elements above principal diagonal are 1.

(3) All the remaining elements are zero.

(4) Matrix B has 'r-1' zero and other elements as 1.

(5) Matrix C has all partial fraction coefficients.

$$\dot{x}_1 = -2x_1 + x_2, \quad \dot{x}_2 = u(t) - 2x_2$$

$$\dot{x}_3 = u(t) - 3x_3(t)$$

$$y(t) = x_1(t) - x_2(t) + x_3(t)$$

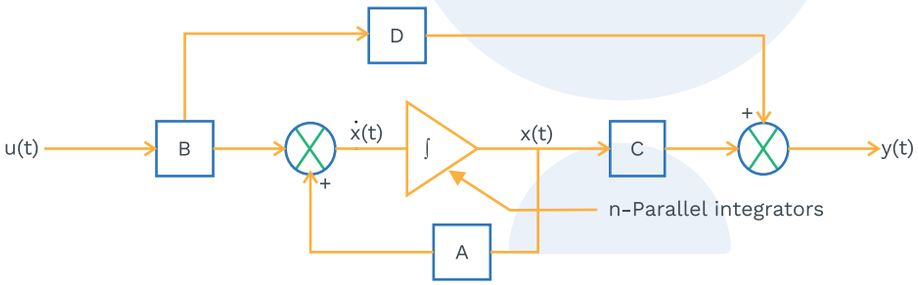
State model-

$$\dot{x} = Ax + Bu \quad \& \quad y = Cx + Du$$

$$A = \begin{bmatrix} -2 & 1 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & -1 & 1 \end{bmatrix}, D = 0$$

State diagram of standard state model



$$\dot{x}(t) = Ax(t) + Bu(t)$$

$$y(t) = Cx(t) + Du(t)$$

Example-5.

For a system with the transfer function $H(s) = \dfrac{4(s-3)}{s^3 + 5s^2 - 3s + 1}$, the matrix B in the state space form $\dot{x} = Ax + Bu$ is equal to-

(a) $\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ (c) $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ (d) $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

Solution: (d)

$$H(s) = \frac{Y(s)}{U(s)} = \frac{4(s-3)}{s^3 + 5s^2 - 3s + 1}$$

$$\frac{Y(s)}{X_1(s)}\frac{X_1(s)}{U(s)} = 4(s-3)\times\left(\frac{1}{s^3+5s^2-3s+1}\right)$$

$$\frac{X_1(s)}{U(s)} = \frac{1}{s^3+5s^2-3s+1}$$

$$s^3X_1(s)+5s^2X_1(s)-3sX_1(s)+X_1(s)=U(s)$$

$$\frac{d^3x_1}{dt^3}+5\frac{d^2x_1}{dt^2}-3\frac{dx_1}{dt}+x_1=u(t)$$

$$\frac{dx_1}{dt}=\dot{x}_1=x_2$$

$$\frac{d^2x_1}{dt^2}=\dot{x}_2=x_3$$

$$\dot{x}_3+5x_3-3x_2+x_1=u(t)$$

$$\dot{x}_3=-x_1+3x_2-5x_3+u(t)$$

$$\dot{x}_1=x_2$$

$$\dot{x}_2=x_3$$

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix}=\begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ -1 & 3 & -5\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}+\begin{bmatrix}0\\ 0\\ 1\end{bmatrix}u(t)$$

Example-6.

A system is described by the differential equation-

$$\frac{d^3y(t)}{dt^3}+8\frac{d^2y(t)}{dt^2}+13\frac{dy(t)}{dt}+10y(t)=8u(t)$$

Where, y(t) is the output and u(t) is the input to the system. Obtain the state space representation of the system.

Solution:

$$\dot{x}_1=x_2,\dot{x}_2=x_3,$$

$$\dot{x}_3=8u(t)-10x_1-13x_2-8x_3$$

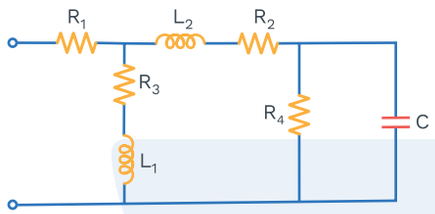
{Consider $y=x_1$, such that $\dot{y}=\dot{x}_2,\dot{y}=\dot{x}_1,\dot{y}=\dot{x}_3$}

In matrix form-

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix}=\begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ -10 & -13 & -8\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}+\begin{bmatrix}0\\ 0\\ 8\end{bmatrix}u(t)$$

Example-7.

The minimum number of states required to describe the network shown in the figure are-



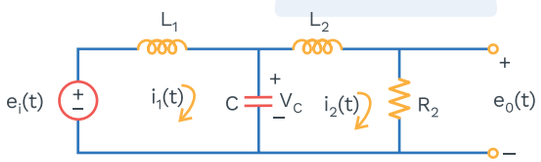
(a) 1 (b) 2 (c) 3 (d) 4

Solution: (c)

The number of states required to describe a network is equal to the number of independent energy storing elements in the network.

There are 3 energy storing elements i.e. L_1, L_2 and C.

Obtain the state model of the given electrical network



u(t) = input = $e_i(t)$

y(t) = output = $e_0(t)$

State variables are-

$$x_1(t)=i_1(t), x_2(t)=i_2(t), x_3(t)=V_C(t)$$

$$e_i(t)=L_1\frac{di_1(t)}{dt}+V_C(t)$$

$$\frac{di_1(t)}{dt}=\frac{1}{L_1}e_i(t)-\frac{1}{L_1}V_C(t)$$

$$\dot{x}_1(t)=\frac{1}{L_1}u(t)-\frac{1}{L_1}x_3(t).....(1)$$

Now, $V_C(t) = L_2 \frac{di_2(t)}{dt} + i_2(t)R_2$

$$\frac{di_2(t)}{dt} = \frac{1}{L_2} V_C(t) - \frac{R_2}{L_2} i_2(t)$$

$$\dot{x}_2(t) = \frac{1}{L_2} x_3(t) - \frac{R_2}{L_2} x_2(t) \ldots\ldots(2)$$

and $C\frac{dV_c(t)}{dt} = i_1(t) - i_2(t)$

$$\frac{dV_C(t)}{dt} = \frac{1}{C} i_1(t) - \frac{1}{C} i_2(t)$$

$$\dot{x}_3(t) = \frac{1}{C} x_1(t) - \frac{1}{C} x_2(t) \ldots\ldots(3)$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 0 & \frac{-1}{L_1} \\ 0 & -\frac{R_2}{L_2} & \frac{1}{L_2} \\ \frac{1}{C} & -\frac{1}{C} & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} \frac{1}{L_1} \\ 0 \\ 0 \end{bmatrix} u(t)$$

i.e. $\dot{x}(t) = A\,x(t) + B\,u(t)$

also $e_0(t) = i_2(t)R_2 \Rightarrow y(t) = R_2 x_2(t)$

$$y(t) = \begin{bmatrix} 0 & R_2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

i.e. $y(t) = C\,x(t) + D\,u(t)$

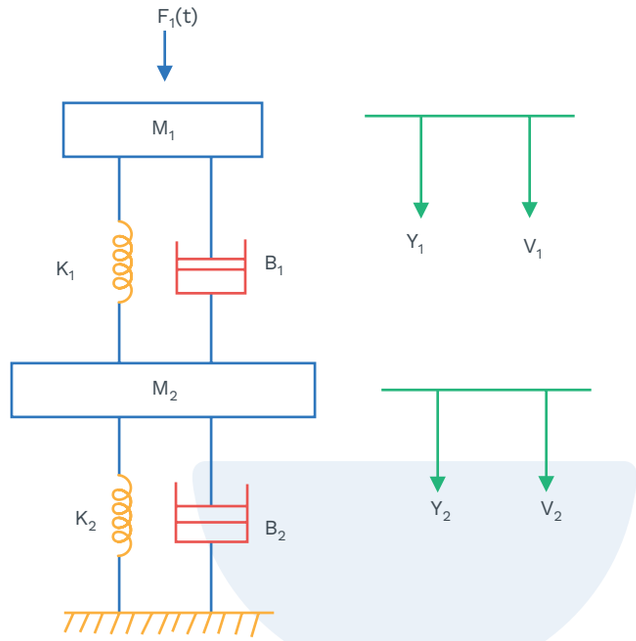
Where, D = [0]

Rack your Brain

Find the number of state variables required if the system consist of five energy storing element.

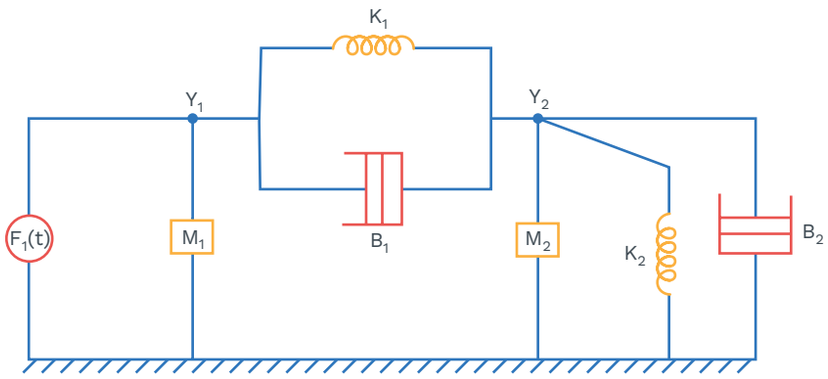
Example-8.

Consider a mechanical system shown in the figure below. For shown displacements and velocities, obtain the state model in standard form. Assume velocity of M_2 as output.



Solution:

$$x_1(t) = Y_1(t), x_3 = V_1(t)$$
$$x_2(t) = Y_2(t), x_4 = V_2(t)$$
$$U(t) = F_1(t), Y(t) = V_2(t)$$



At node Y_1,

$$F_1 = M_1 \frac{d^2Y_1}{dt^2} + K_1(Y_1 - Y_2) + \frac{B_1 d(Y_1 - Y_2)}{dt} \quad \text{.....(1)}$$

At node Y_2,

$$0 = M_2\frac{d^2Y_2}{dt^2} + K_2Y_2 + B_2\frac{dY_2}{dt} + K_1\left(Y_2 - Y_1\right) + \frac{B_1 d\left(Y_2 - Y_1\right)}{dt} \text{.....(2)}$$

Substituting all values in the terms of state variables, we get-

$$Y_1 = x_1, \quad \frac{dY_1}{dt} = \dot{x}_1 = V_1 = x_3, \quad \frac{d^2Y_1}{dt^2} = \dot{x}_3$$

$$Y_2 = x_2, \quad \frac{dY_2}{dt} = \dot{x}_2 = V_2 = x_4, \quad \frac{d^2Y_2}{dt^2} = \dot{x}_4$$

$\therefore$ Substituting in equation (1) and equation (2), we get-

$$U(t) = M_1\dot{x}_3 + K_1\left[x_1 - x_2\right] + B_1\left[x_3 - x_4\right] \text{.....}(3)$$

$$0 = M_2\dot{x}_4 + K_2x_2 + B_2x_4 + K_1\left[x_2 - x_1\right] + B_1\left[x_4 - x_3\right] \text{.....(4)}$$

From equation (3) and (4) we can write-

$$\dot{x}_3 = \frac{1}{M_1}\left[U(t) - K_1\left(x_1 - x_2\right) - B_1\left(x_3 - x_4\right)\right] \text{.....(5)}$$

And $\dot{x}_4 = \frac{1}{M_2}\left[-K_2x_2 - B_2x_4 - K_1\left(x_2 - x_1\right) - B_1\left(x_4 - x_3\right)\right]$.....(6)

And $\dot{x}_1 = x_3, \dot{x}_2 = x_4$, $Y(t) = V_2(t) = x_4(t)$

$\therefore$ State model can be constructed in the standard form-

$$\dot{X} = AX + BU$$

$$Y = CX + DU$$

Where $A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -\frac{K_1}{M_1} & +\frac{K_1}{M_1} & -\frac{B_1}{M_1} & +\frac{B_1}{M_1} \\ \frac{K_1}{M_2} & \frac{-(K_1+K_2)}{M_2} & \frac{B_1}{M_2} & \frac{-(B_1+B_2)}{M_2} \end{bmatrix}$

$$B = \begin{bmatrix} 0 \\ 0 \\ \frac{1}{M_1} \\ 0 \end{bmatrix}, C = \begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix}$$

Example-9.

For a system having-

$$G(s) = \frac{Y(s)}{U(s)} = \frac{s^2 + 4s + 1}{s^3 + 3s^2 + 7s + 9}$$

Obtain the state model.

Solution:

$$\frac{Y(s)}{U(s)} = \frac{s^2 + 4s + 1}{s^3 + 3s^2 + 7s + 9} \qquad \ldots (1)$$

Multiplying and dividing $X_1(s)$ in equation (1)-

$$\frac{Y(s)}{X_1(s)}\frac{X_1(s)}{U(s)} = \left(s^2 + 4s + 1\right)\frac{1}{s^3 + 3s^2 + 7s + 9}$$

On comparing, we get:

$$\frac{X_1(s)}{U(s)} = \frac{1}{s^3 + 3s^2 + 7s + 9}$$

$$s^3X_1(s) + 3s^2X_1(s) + 7sX_1(s) + 9X_1(s) = U(s)$$

$$\frac{d^3x_1}{dt^3} + 3\frac{d^2x_1}{dt^2} + 7\frac{dx_1}{dt} + 9x_1 = u(t)$$

$$\frac{dx_1}{dt} = x_2 = \dot{x}_1, \frac{d^2x_1}{dt^2} = x_3 = \dot{x}_2, \frac{d^3x_1}{dt^3} = \dot{x}_3$$

Then,

$$\dot{x}_3 = u(t) - 9x_1 - 7x_2 - 3x_3$$

$$\Rightarrow \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -9 & -7 & -3 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}u(t)$$

Now,

$$Y(s) = s^2X_1(s) + 4sX_1(s) + X_1(s)$$

$$y = \frac{d^2x_1}{dt^2} + 4\frac{dx_1}{dt} + x_1$$

$$y = \begin{bmatrix} 1 & 4 & 1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Example-10.
A certain linear time invariant system has the state and the output equations which are given below.

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix}=\begin{bmatrix}3 & -2\\ 0 & 4\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}+\begin{bmatrix}0\\ 3\end{bmatrix}u$$

$$y=\begin{bmatrix}2 & 3\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}$$

If $x_1(0)=3, x_2(0)=-2, u(0)=0$ then $\left.\frac{dy}{dt}\right|_{t=0}$ is –

Solution:

$$\dot{x}_1=3x_1-2x_2$$

$$\dot{x}_1(0)=3x_1(0)-2x_2(0)$$

$$=9+4=13$$

$$\dot{x}_2=4x_2+3u$$

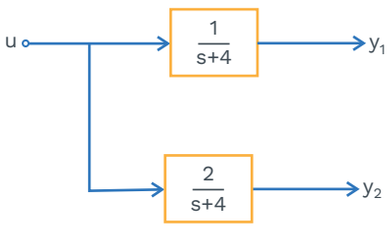
$$\dot{x}_2(0)=4x_2(0)+3u(0)=-8$$

$$y=2x_1+3x_2$$

$$\frac{dy}{dt}=2\dot{x}_1+3\dot{x}_2$$

$$\left.\frac{dy}{dt}\right|_{t=0}=2\dot{x}_1(0)+3\dot{x}_2(0)=26-24=2$$

Example-11.
The block diagram of a system with one input as u and two outputs as y_1 and y_2 is given below.



Find the state space model as well as the output vector $y=\begin{bmatrix}y_1 & y_2\end{bmatrix}^T$.

Solution:

$$\frac{Y_1(s)}{U(s)} = \frac{1}{s+4}; \frac{Y_2(s)}{U(s)} = \frac{2}{s+4}$$

$$\frac{Y_1(s)}{X_1(s)}\frac{X_1(s)}{U(s)} = \frac{1}{s+4}; \frac{Y_2(s)}{X_2(s)}\frac{X_2(s)}{U(s)} = \frac{2}{s+4}$$

$$\frac{X_1(s)}{U(s)} = \frac{1}{s+4} \text{ and } \frac{Y_1(s)}{X_1(s)} = 1; \quad \frac{X_2(s)}{U(s)} = \frac{1}{s+4} \text{ and } \frac{Y_2(s)}{X_2(s)} = 2$$

$$sX_1(s) + 4X_1(s) = U(s); \; sX_2(s) + 4X_2(s) = U(s)$$

$$Y_1(s) = X_1(s); \quad Y_2(s) = 2X_2(s)$$

$$\dot{x}_1 + 4x_1 = u(t); \; \dot{x}_2 + 4x_2 = u(t)$$

$$y_1(t) = x_1 \; ; \; y_2 = 2x_2$$

From the above equations, we get-

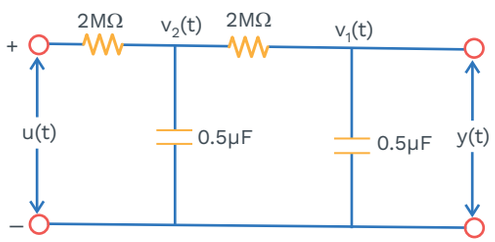
$$\dot{x}_1 = -4x_1 + u(t); \; \dot{x}_2 = -4x_2 + u(t)$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -4 & 0 \\ 0 & -4 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u(t)$$

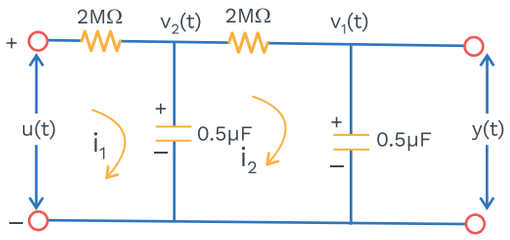
$$y = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Example-12.

For the system shown in the figure below, obtain the state model choosing $v_1(t)$ and $v_2(t)$ as the state variables.



Solution:



$$i_1(t) = \frac{u(t) - v_2(t)}{2 \times 10^6} \text{.....(1)}$$

$$v_2(t) = \frac{1}{C}\int (i_1 - i_2)\,dt$$

$$\text{i.e. } i_1 - i_2 = C\frac{dv_2(t)}{dt} \text{.....(2)}$$

$$i_2(t) = \frac{v_2(t) - v_1(t)}{2 \times 10^6} \text{.....(3)}$$

and

$$v_1(t) = \frac{1}{C}\int i_2 dt$$

$$\therefore i_2(t) = C\frac{dv_1(t)}{dt} \text{.....(4)}$$

Substituting (1) and (3) in 2, we get-

$$C\frac{dv_2(t)}{dt} = \left[\frac{u(t) - v_2(t)}{2 \times 10^6}\right] - \left[\frac{v_2(t) - v_1(t)}{2 \times 10^6}\right]$$

$$\therefore \frac{dv_2(t)}{dt} = \frac{1}{C}\left[\frac{u(t)}{2 \times 10^6}\right] + \frac{1}{C}\frac{v_1(t)}{2 \times 10^6} - \frac{1}{C}\left[\frac{2v_2(t)}{2 \times 10^6}\right]$$

$$C = 0.5 \times 10^{-6} F$$

$$\therefore \frac{dv_2}{dt} = v_1(t) - 2v_2(t) + u(t) \text{.....(5)}$$

Using (3) in (4),

$$C\frac{dv_1(t)}{dt} = \frac{v_2(t) - v_1(t)}{2 \times 10^6}$$

$$\therefore \frac{dv_1(t)}{dt} = -v_1(t) + v_2(t) \text{.....(6)}$$

Select $x_1(t) = v_1(t)$ and $x_2(t) = v_2(t)$ using selected state variables,

$$\dot{x}_1 = -x_1 + x_2$$

and $\dot{x}_2 = x_1 - 2x_2 + u(t)$

and $y(t) = v_1(t) = x_1(t)$

Hence, the state model is-

$$\dot{x}(t) = Ax(t) + Bu(t)$$

$$y(t) = Cx(t) + Du(t)$$

$$A = \begin{bmatrix} -1 & +1 \\ 1 & -2 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, C = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

$D = 0$

Transfer function from state model

$$\dot{x}(t) = Ax(t) + Bu(t)$$

$$y(t) = Cx(t) + Du(t)$$

Taking Laplace transform on both sides,

$$[sX(s) - x(0)] = AX(s) + BU(s)$$

$$Y(s) = CX(s) + DU(s) \ldots\ldots\ldots\ldots (i)$$

Transfer function is based on the assumption of zero initial conditions i.e. $x(0) = 0$

$$\therefore sX(s) = AX(s) + BU(s)$$

$$sX(s) - AX(s) = BU(s)$$

$$sIX(s) - AX(s) = BU(s)$$

$$(sI - A)X(s) = BU(s)$$

Multiplying both sides by $[sI - A]^{-1}$ -

$$(sI - A)^{-1}(sI - A)X(s) = (sI - A)^{-1}BU(s)$$

$$X(s) = [sI - A]^{-1}BU(s) \ldots\ldots (ii)$$

Put the value (ii) in (i)-

$$Y(s) = C[sI - A]^{-1}BU(s) + DU(s)$$

$$Y(s) = \left[C[sI - A]^{-1}B + D\right]U(s)$$

Previous Year's Questions

The transfer function for the state variable representation $\dot{X} = AX + BU, Y = CX + DU,$ is given by

(a) $D + C(sI - A)^{-1}B$

(b) $B(sI - A)^{-1}C + D$

(c) $D(sI - A)^{-1}B + C$

(d) $C(sI - A)^{-1}D + B$

Ans (a)

Hence the transfer function is,

$$T(s) = \frac{Y(s)}{U(s)} = C\left[sI - A\right]^{-1} B + D$$

Where, $\left[sI - A\right]^{-1} = \frac{Adj\left[sI - A\right]}{|sI - A|}$

Example-13.
Find the transfer function for the following system.

$$\dot{X} = \begin{bmatrix} 0 & 1 \\ -4 & -6 \end{bmatrix} X + \begin{bmatrix} 0 \\ 2 \end{bmatrix} U$$

$$Y = [0 \quad 3]X + [0]\, U$$

Solution:

$$A = \begin{bmatrix} 0 & 1 \\ -4 & -6 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 2 \end{bmatrix}, C = [0,\ 3], D = [0]$$

$$[sI - A] = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ -4 & -6 \end{bmatrix}$$

$$= \begin{bmatrix} s & -1 \\ 4 & s+6 \end{bmatrix}$$

$$\left[sI - A\right]^{-1} = \frac{1}{s(s+6)+4} \begin{bmatrix} s+6 & 1 \\ -4 & s \end{bmatrix}$$

Transfer function, $T(s) = C\left[sI - A\right]^{-1} B + D$

$$= \frac{1}{s(s+6)+4} \left[[0 \quad 3] \begin{bmatrix} s+6 & 1 \\ -4 & s \end{bmatrix} \begin{bmatrix} 0 \\ 2 \end{bmatrix} \right] + [0]$$

$$= \frac{1}{\left[s(s+6)+4\right]} [-12 \quad 3s] \begin{bmatrix} 0 \\ 2 \end{bmatrix} + [0]$$

$$T(s) = \frac{6s}{s(s+6)+4} = \frac{6s}{s^2 + 6s + 4}$$

Example-14.
Find the transform of the system having state model-

$$\dot{X} = \begin{bmatrix} -5 & 1 \\ -4 & 0 \end{bmatrix} X + \begin{bmatrix} 0 \\ 1 \end{bmatrix} U \text{ and } Y = [1 \quad 0]X$$

Previous Year's Questions

A system is described by the state equation $\dot{X} = AX + BU$, the output is given by $Y = CX$

Where, $A = \begin{pmatrix} -4 & -1 \\ 3 & -1 \end{pmatrix}$, $B = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$C = [1 \quad 0]$

Transfer function G(s) of the system is

(a) $\frac{s}{s^2 + 5s + 7}$ (b) $\frac{1}{s^2 + 5s + 7}$

(c) $\frac{s}{s^2 + 3s + 2}$ (d) $\frac{1}{s^2 + 3s + 2}$

Ans (a)

Solution:

$$A=\begin{bmatrix}-5 & 1\\ -4 & 0\end{bmatrix}\ B=\begin{bmatrix}0\\ 1\end{bmatrix}\ C=\begin{bmatrix}1 & 0\end{bmatrix}, D=0$$

$$T(s)=\frac{Y(s)}{U(s)}=C\left[sI-A\right]^{-1}B+D$$

$$\left[sI-A\right]=s\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}-\begin{bmatrix}-5 & 1\\ -4 & 0\end{bmatrix}=\begin{bmatrix}s+5 & -1\\ 4 & s\end{bmatrix}$$

$$Adj\left[sI-A\right]=\begin{bmatrix}s & 1\\ -4 & s+5\end{bmatrix}$$

$$|\,sI-A\,|=s(s+5)+4=s^2+5s+4=(s+4)(s+1)$$

$$T(s)=\frac{C\ Adj\ [sI-A]B}{|\,sI-A\,|}=\frac{\begin{bmatrix}1 & 0\end{bmatrix}\begin{bmatrix}s & 1\\ -4 & s+5\end{bmatrix}\begin{bmatrix}0\\ 1\end{bmatrix}}{(s+4)(s+1)}$$

$$T(s)=\frac{1}{(s+4)(s+1)}$$

Example-15.

A linear time invariant single input single output system has a state space model given by-

$$\frac{dx}{dt}=Ax+Bu, y=Cx$$

Where, $A=\begin{bmatrix}0 & 1\\ -6 & -3\end{bmatrix}, B=\begin{bmatrix}0\\ 1\end{bmatrix}, C=\begin{bmatrix}1 & 0\end{bmatrix}$

x = state vector, u = input, y = output.
Find out the damping ratio.

Solution:

$$T(s)=C\left[sI-A\right]^{-1}B$$

$$A=\begin{bmatrix}0 & 1\\ -6 & -3\end{bmatrix}B=\begin{bmatrix}0\\ 1\end{bmatrix}\text{ and }C=\begin{bmatrix}1 & 0\end{bmatrix}$$

$$T(s)=\begin{bmatrix}1 & 0\end{bmatrix}\begin{bmatrix}s & -1\\ 6 & s+3\end{bmatrix}^{-1}\begin{bmatrix}0\\ 1\end{bmatrix}$$

$$T(s) = \frac{\begin{bmatrix}1 & 0\end{bmatrix}\begin{bmatrix}(s+3) & 1\\ -6 & s\end{bmatrix}\begin{bmatrix}0\\1\end{bmatrix}}{s^2+3s+6} = \frac{1}{s^2+3s+6}$$

From the transfer function,

$$2\xi\omega_n = 3$$

$$\omega_n = \sqrt{6}$$

$$\xi = \frac{3}{2\sqrt{6}} = \frac{\sqrt{3}}{\sqrt{8}} = \sqrt{\frac{3}{8}}$$

$$\xi = \sqrt{\frac{3}{8}} = 0.6$$

MIMO SYSTEM [Multiple Inputs Multiple Outputs]

For MIMO system, instead of transfer function there exist transfer matrix. In MIMO system there exist a relationship between each output and all the inputs.

Example-16.

The state matrix is given below.

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}0 & 4\\ -2 & -6\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix}\begin{bmatrix}u_1\\ u_2\end{bmatrix}$$

$$\begin{bmatrix}y_1\\ y_2\end{bmatrix} = \begin{bmatrix}2 & 1\\ 1 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}$$

Find the output matrix.

Solution:

$$A = \begin{bmatrix}0 & 4\\ -2 & -6\end{bmatrix}, B = \begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix}, C = \begin{bmatrix}2 & 1\\ 1 & 0\end{bmatrix}$$

$$D = \begin{bmatrix}0\end{bmatrix}$$

$$\text{Transfer Matrix} = C\left[sI - A\right]^{-1}B + D$$

$$\left[sI - A\right] = \begin{bmatrix}s & -4\\ 2 & s+6\end{bmatrix}$$

$$\left[sI - A\right]^{-1} = \frac{\begin{bmatrix} s+6 & 4 \\ -2 & s \end{bmatrix}}{(s+2)(s+4)}$$

$$T.M. = C\left[sI - A\right]^{-1}B = \frac{\begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} s+6 & 4 \\ -2 & s \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}}{(s+2)(s+4)}$$

$$= \frac{\begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} s+10 & s+10 \\ s-2 & s-2 \end{bmatrix}}{(s+2)(s+4)} = \frac{\begin{bmatrix} 3s+18 & 3s+18 \\ s+10 & s+10 \end{bmatrix}}{(s+2)(s+4)}$$

$$= \begin{bmatrix} \frac{3s+18}{(s+2)(s+4)} & \frac{3s+18}{(s+2)(s+4)} \\ \frac{s+10}{(s+2)(s+4)} & \frac{s+10}{(s+2)(s+4)} \end{bmatrix}$$

$Y(s) = T.M.\,U(s)$; (T.M. = transfer matrix)

$$\begin{bmatrix} Y_1(s) \\ Y_2(s) \end{bmatrix} = \begin{bmatrix} \frac{3s+18}{(s+2)(s+4)} & \frac{3s+18}{(s+2)(s+4)} \\ \frac{s+10}{(s+2)(s+4)} & \frac{s+10}{(s+2)(s+4)} \end{bmatrix}\begin{bmatrix} U_1(s) \\ U_2(s) \end{bmatrix}$$

Eigen Values, Eigen vector and Model matrix (M)

Eigen values: Eigen values are the closed loop poles of the system.

Consider an equation, AX = Y

If vector X exist such that A transforms it to a vector λX, then X is called solution equation.

$$AX = \lambda X$$

i.e. $\lambda X - AX = 0$

i.e. $\left[\lambda I - A\right]X = 0$

$|\lambda I - A| \Rightarrow$ characteristic polynomial and $|\lambda I - A| = 0 \Rightarrow$ characteristic equation. After expanding-

$$|\lambda I - A| = \lambda^n + a_1\lambda^{n-1} + ... + a_n = 0$$

The value of λ that satisfies the above equation is known as eigen value.

Stability from state model

Transfer function,

$$T(s) = \frac{Y(s)}{U(s)} = C\left[sI - A\right]^{-1} B + D$$

$$= \frac{C\,\text{adj}\left[sI - A\right]B + \left|sI - A\right|D}{\left|sI - A\right|}$$

Denominator of closed loop transfer function represents the characteristic equation.

$$\left|sI - A\right| = 0$$

The roots of characteristic equation are the poles of closed loop transfer function. So, for the system to be stable, all roots must be negative.

Example-17.

For the given state model-

$$\dot{x} = \begin{bmatrix} 1 & -4 \\ 3 & -6 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad u, y = \begin{bmatrix} 1 & 1 \end{bmatrix} x$$

Determine the stability of the given model.

Solution:

To determine stability, we must find eigen values.

$$A = \begin{bmatrix} 1 & -4 \\ 3 & -6 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, C = \begin{bmatrix} 1 & 1 \end{bmatrix}$$

$$\left|\lambda I - A\right| = \left|\lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & -4 \\ 3 & -6 \end{bmatrix}\right|$$

$$= \begin{vmatrix} \lambda - 1 & 4 \\ -3 & \lambda + 6 \end{vmatrix}$$

$\therefore (\lambda - 1)(\lambda + 6) + 12 = 0$, i.e. $\lambda^2 + 5\lambda + 6 = 0$

So, $\lambda_1 = -2, \lambda_2 = -3$

Both the eigen values are negative. So, the system is stable.

Eigen vector: Any non-zero vector X_i such that $AX_i = \lambda_i X_i$ is said to be eigen vector when It satisfies $(\lambda_i I - A) X_i = 0$.

It is denoted by M_i.

Model matrix- $\lambda_1, \lambda_2 \ldots\ldots \lambda_n$ be eigen values and $M_1, M_2, \ldots\ldots M_n$ be eigen vectors. By placing all the eigen vectors corresponding to eigen values in matrix form, then the matrix obtained is called model matrix.

$$M = \text{Model Matrix} = \left[M_1 \; M_2 \ldots\ldots M_n\right]$$

$$M^{-1}AM = \wedge \text{ (Diagonal Matrix)}$$

Example-18.

Obtain the eigen values, eigen vectors and model matrix for the matrix–

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix}$$

Solution:

Eigen value $\Rightarrow$ Roots of $|\lambda I - A| = 0$

$$|\lambda I - A| = \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & +1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} = 0$$

$$\begin{vmatrix} \lambda & -1 & 0 \\ 0 & \lambda & -1 \\ 6 & 11 & \lambda + 6 \end{vmatrix} = 0$$

$$\lambda^3 + 6\lambda^2 + 11\lambda + 6 = 0$$

Now, test $\lambda = -1$ for its root.

$$\begin{array}{c|cccc} -1 & 1 & 6 & 11 & 6 \\ & & -1 & -5 & -6 \\ \hline & 1 & 5 & 6 & 0 \end{array}$$

Hence, $\lambda = -1$ is a root.

$$(\lambda + 1)(\lambda^2 + 5\lambda + 6) = 0$$

$$(\lambda + 1)(\lambda + 2)(\lambda + 3) = 0$$

$\lambda_1 = -1,\ \lambda_2 = -2,\ \lambda_3 = -3$ are eigen values.

Eigen vector-

$$\lambda = \lambda_1 = -1$$

Gray Matter Alert!!!

The roots of $|sI - A| = 0$ are also called eigen values.

$$\therefore \left[\lambda_1 I - A\right] = \begin{bmatrix} -1 & -1 & 0 \\ 0 & -1 & -1 \\ 6 & 11 & 5 \end{bmatrix}$$

$$M_1 = \begin{bmatrix} C_{11} \\ C_{12} \\ C_{13} \end{bmatrix} \text{ ; where C = Cofactor}$$

$$M_1 = \begin{bmatrix} 6 \\ -6 \\ 6 \end{bmatrix} \text{ ; taking out common factor}$$

$$M_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

$$\lambda = \lambda_2 = -2$$

$$\left[\lambda_2 I - A\right] = \begin{bmatrix} -2 & -1 & 0 \\ 0 & -2 & -1 \\ 6 & 11 & 4 \end{bmatrix}$$

$$M_2 = \begin{bmatrix} C_{11} \\ C_{12} \\ C_{13} \end{bmatrix} = \begin{bmatrix} 3 \\ -6 \\ 12 \end{bmatrix} \text{ ; taking out common factor}$$

$$M_2 = \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix}$$

$$\lambda_3 = -3$$

$$\left[\lambda_3 I - A\right] = \begin{bmatrix} -3 & -1 & 0 \\ 0 & -3 & -1 \\ 6 & 11 & 3 \end{bmatrix}$$

$$M_3 = \begin{bmatrix} C_{11} \\ C_{12} \\ C_{13} \end{bmatrix} = \begin{bmatrix} 2 \\ -6 \\ 18 \end{bmatrix} \text{ ; taking out common factor}$$

$$M_3 = \begin{bmatrix} 1 \\ -3 \\ 9 \end{bmatrix}$$

Gray Matter Alert!!!

(1) Both A & $\wedge$ have same characteristic equation.

(2) Both A & $\wedge$ have same eigen values.

Hence, due to the transformation $M^{-1}AM$, eigen values of matrix A remains unchanged.

Therefore, the model matrix:

$$M = \begin{bmatrix} M_1 & M_2 & M_3 \end{bmatrix}$$

$$M = \begin{bmatrix} 1 & 1 & 1 \\ -1 & -2 & -3 \\ 1 & 4 & 9 \end{bmatrix}$$

Solution of State Equation

Consider a non-homogeneous equation-

$$\dot{x}(t) = A\,x(t) + B\,u(t)$$

$$\dot{x}(t) - A\,x(t) = B\,u(t)$$

Multiply e^{-At} on both sides-

$$e^{-At}\left[\dot{x}(t) - A\,x(t)\right] = e^{-At}\,B\,u(t)$$

$$\left\{\text{But } e^{-At}\,\dot{x}(t) - e^{-At}\,Ax(t) = \frac{d}{dt}e^{-At}\,x(t)\right\}$$

$$\frac{d}{dt}\left[e^{-At}x(t)\right] = e^{-At}\,B\,u(t)$$

Integrating both sides with lim (0 to t)-

$$e^{-At}x(t)\Big|_0^t = \int_0^t e^{-A\tau}\,B\,u(\tau)\,d\tau$$

$$e^{-At}\,x(t) - x(0) = \int_0^t e^{-A\tau}\,B\,u(\tau)\,d\tau$$

Now, again multiplying e^{At} on both sides, we get-

$$x(t) - e^{At}x(0) = \int_0^t e^{At}e^{-A\tau}B\,u(\tau)\,d\tau$$

$$x(t) = e^{At}x(0) + \int_0^t e^{A(t-\tau)}B\,u(\tau)\,d\tau$$

This equation is the solution to non-homogeneous equation.

The solution is divided into two parts-

(i) **Zero input response:** The response of state equation under zero input is known as zero input response (ZIR). It is the first part i.e. $e^{At}x(0)$.

(ii) **Zero state solution:** The second part i.e. $\int_0^t e^{A(t-\tau)}Bu(\tau)d\tau$ is called as forced solution of zero state response.

Solution of state equation by Laplace transform

To convert integral-differential equation to simple algebraic equation, we use Laplace transform method.

$$\dot{x}(t) = Ax(t) + Bu(t)$$

Taking Laplace transform-

$$sX(s) - x(0) = AX(s) + BU(s)$$

$$sX(s) - AX(s) = x(0) + BU(s)$$

$$[sI - A]X(s) = x(0) + BU(s)$$

Multiplying $[sI - A]^{-1}$ on both sides-

$$X(s) = [sI - A]^{-1}x(0) + [sI - A]^{-1}BU(s)$$

Applying inverse Laplace transform to the above equation, we get-

$x(t) = L^{-1}(sI\text{-}A)^{-1}\, x(0) + L^{-1}\,(sI\text{-}A)^{-1}\, BU(s)$

$= e^{At}\, x(0) + L^{-1}\, \phi(s)\, BU(s)$

$x(t) = \phi(t)\, x(0) + L^{-1}\, \phi(s)\, BU(s)$

$= ZIR + ZSR$

Therefore, we can say that-

$$L^{-1}[sI - A]^{-1} = \phi(t) = e^{At} \rightarrow ZIR$$

$$L^{-1}[\phi(s)BU(s)] = \int_0^t \phi(t-\tau)Bu(\tau)d\tau \rightarrow ZSR$$

Previous Year's Questions

A system is described by the following state and output equations

$$\frac{dx_1(t)}{dt} = -3x_1(t) + x_2(t) + 2u(t)$$

$$\frac{dx_2(t)}{dt} = -2x_2(t) + u(t)$$

$y(t) = x_1(t)$ when u(t) is the input and y (t) is the output. The state-transition matrix of the above system is

(a) $\begin{pmatrix} e^{-3t} & 0 \\ e^{-2t} + e^{-3t} & e^{-2t} \end{pmatrix}$

(b) $\begin{pmatrix} e^{-3t} & e^{-2t} - e^{-3t} \\ 0 & e^{-2t} \end{pmatrix}$

(c) $\begin{pmatrix} e^{-3t} & e^{-2t} + e^{-3t} \\ 0 & e^{-2t} \end{pmatrix}$

(d) $\begin{pmatrix} e^{3t} & e^{-2t} - e^{-3t} \\ 0 & e^{-2t} \end{pmatrix}$

Ans (b)

Properties of state transition matrix

$\phi(t) = e^{At}$ = State transition matrix

(1) $\phi(0) = e^{A(0)} = I$, Identity matrix

(2) $\phi(t) = e^{At} = \left(e^{-At}\right)^{-1} = \left[\phi(-t)\right]^{-1}$

$\Rightarrow \phi^{-1}(t) = \phi(-t)$

(3) $\phi(t_1 + t_2) = e^{A(t_1+t_2)} = e^{At_1}.e^{At_2}$

$= \phi(t_1)\phi(t_2) = \phi(t_2)\phi(t_1)$

(4) $e^{A(t+s)} = e^{At}e^{As}$

(5) $e^{(A+B)t} = e^{At}e^{Bt}$ {Only if $AB = BA$}

(6) $\left[\phi(t)\right]^n = \left[e^{At}\right]^n = e^{Ant} = \phi(nt)$

(7) $\phi(t_2 - t_1).\phi(t_1 - t_0) = \phi(t_2 - t_0)$

(8) $\phi(t)$ is non-singular matrix for all finite values of t.

Rack your Brain

What is the value of state transition matrix at t = 0?

Example-19.

Obtain the state transition matrix $\phi(t)$ of the following-

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}0 & 1\\ -4 & -5\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}$$

Also find the inverse of the state transition matrix.

Solution:

For the system,

$$A = \begin{bmatrix}0 & 1\\ -4 & -5\end{bmatrix}$$

State transition equation-

$$\phi(t) = e^{At} = L^{-1}\left[sI - A\right]^{-1}$$

Since,

$$\left[sI - A\right] = \begin{bmatrix}s & 0\\ 0 & s\end{bmatrix} - \begin{bmatrix}0 & 1\\ -4 & -5\end{bmatrix} = \begin{bmatrix}s & -1\\ 4 & s+5\end{bmatrix}$$

The inverse of $\left[sI - A\right]$ is given by-

$$\left[sI-A\right]^{-1}=\frac{1}{s^2+5s+4}\begin{bmatrix}s+5 & 1\\ -4 & s\end{bmatrix}=\frac{1}{(s+4)(s+1)}\begin{bmatrix}s+5 & 1\\ -4 & s\end{bmatrix}$$

$$=\begin{bmatrix}\frac{s+5}{(s+4)(s+1)} & \frac{1}{(s+4)(s+1)}\\ \frac{-4}{(s+4)(s+1)} & \frac{s}{(s+4)(s+1)}\end{bmatrix}$$

$$\phi(t)=e^{At}=L^{-1}\left[\left[sI-A\right]^{-1}\right]$$

$$=\begin{bmatrix}\frac{-1}{3}e^{-4t}+\frac{4}{3}e^{-t} & \frac{-1}{3}e^{-4t}+\frac{1}{3}e^{-t}\\ \frac{4}{3}e^{-4t}-\frac{4}{3}e^{-t} & \frac{4}{3}e^{-4t}-\frac{1}{3}e^{-t}\end{bmatrix}$$

Using $\phi^{-1}(t)=\phi(-t)$

$$\phi^{-1}(t)=e^{-At}=\begin{bmatrix}\frac{-1}{3}e^{4t}+\frac{4}{3}e^{t} & \frac{-1}{3}e^{4t}+\frac{1}{3}e^{t}\\ \frac{4}{3}e^{4t}-\frac{4}{3}e^{t} & \frac{4}{3}e^{4t}-\frac{1}{3}e^{t}\end{bmatrix}$$

Example-20.

For the system given below, obtain-

(i) Zero input response (ii) Zero state response

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix}=\begin{bmatrix}2 & 0\\ 2 & 2\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}+\begin{bmatrix}1\\ 1\end{bmatrix}u(t)$$

Where u(t) is the unit step occurring at t = 0

and $x^T(0)=[1 \quad 0]$

Solution:

$$A=\begin{bmatrix}2 & 0\\ 2 & 2\end{bmatrix}$$

$$\left[sI-A\right]=s\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}-\begin{bmatrix}2 & 0\\ 2 & 2\end{bmatrix}=\begin{bmatrix}s-2 & 0\\ -2 & s-2\end{bmatrix}$$

$$\left[sI-A\right]^{-1}=\frac{\begin{bmatrix}s-2 & 0\\ 2 & s-2\end{bmatrix}}{(s-2)^2}=\begin{bmatrix}\frac{1}{s-2} & 0\\ \frac{2}{(s-2)^2} & \frac{1}{s-2}\end{bmatrix}$$

$$e^{At} = L^{-1}\left[sI - A\right]^{-1} = L^{-1}\begin{bmatrix} \frac{1}{s-2} & 0 \\ \frac{2}{(s-2)^2} & \frac{1}{s-2} \end{bmatrix}$$

$$= \begin{bmatrix} e^{2t} & 0 \\ 2te^{2t} & e^{2t} \end{bmatrix}$$

$x(t) = e^{At}x(0) = \text{Zero input response}$

$$= \begin{bmatrix} e^{2t} & 0 \\ 2te^{2t} & e^{2t} \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} e^{2t} \\ 2te^{2t} \end{bmatrix}$$

$$ZIR = \begin{bmatrix} e^{2t} \\ 2te^{2t} \end{bmatrix}$$

Now, ZSR = $x(t) = L^{-1}\left\{\phi(s)BU(s)\right\}$

Where,

$$\phi(s) = L\left\{\phi(t)\right\} = L\left\{e^{At}\right\} = \left[sI - A\right]^{-1}$$

So, from above-

$$\phi(s) = \begin{bmatrix} \frac{1}{s-2} & 0 \\ \frac{2}{(s-2)^2} & \left(\frac{1}{s-2}\right) \end{bmatrix}$$

$B = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $U(s) = \left(\frac{1}{s}\right)$; as input is a unit step

$$\phi(s)BU(s) = \begin{bmatrix} \frac{1}{s-2} & 0 \\ \frac{2}{(s-2)^2} & \frac{1}{s-2} \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix}\left[\frac{1}{s}\right]$$

$$= \begin{bmatrix} \frac{1}{s-2} \\ \frac{2}{(s-2)^2} + \frac{1}{s-2} \end{bmatrix}\left[\frac{1}{s}\right] = \begin{bmatrix} \frac{1}{s(s-2)} \\ \frac{1}{s}\left[\frac{2}{(s-2)^2} + \frac{1}{s-2}\right] \end{bmatrix}$$

Previous Year's Questions

For the system

$$\dot{x} = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}X + \begin{bmatrix} 1 \\ 1 \end{bmatrix}u; y = \begin{bmatrix} 4 & 0 \end{bmatrix}X,$$

with u as unit impulse and with zero initial state, the output, y, becomes

(a) $2e^{2t}$ (b) $4e^{2t}$

(c) $2e^{4t}$ (d) $4e^{4t}$

Ans (b)

By using partial fraction-

$$= \begin{bmatrix} \frac{\frac{1}{2}}{s-2} - \frac{\frac{1}{2}}{s} \\ \frac{1}{(s-2)^2} \end{bmatrix}$$

$$ZSR = L^{-1}\left[\phi(s)BU(s)\right] = L^{-1}\begin{bmatrix} \frac{\frac{1}{2}}{s-2} - \frac{\frac{1}{2}}{s} \\ \frac{1}{(s-2)^2} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{2}\left(e^{2t}-1\right) \\ te^{2t} \end{bmatrix}$$

Total response,

$$x(t) = ZIR + ZSR$$

$$= \begin{bmatrix} e^{2t} \\ 2te^{2t} \end{bmatrix} + \begin{bmatrix} \frac{1}{2}\left(e^{2t}-1\right) \\ te^{2t} \end{bmatrix} = \begin{bmatrix} \frac{1}{2}\left(3e^{2t}-1\right) \\ 3te^{2t} \end{bmatrix}$$

Example-21.

For the given homogeneous state space equation $\dot{x} = \begin{bmatrix} -4 & 1 \\ 0 & -5 \end{bmatrix} x$. The steady state value $X_{ss} = \lim_{t\to\infty} x(t)$, given the initial state value of $x(0) = \begin{bmatrix} 5 & -5 \end{bmatrix}^T$ is__.

Solution:

Given, $A = \begin{bmatrix} -4 & 1 \\ 0 & -5 \end{bmatrix}$

Now, $\left[sI - A\right] = \begin{bmatrix} s+4 & -1 \\ 0 & s+5 \end{bmatrix}$

$$\left[sI - A\right]^{-1} = \frac{1}{(s+4)(s+5)}\begin{bmatrix} s+5 & 1 \\ 0 & s+4 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{s+4} & \frac{1}{(s+4)(s+5)} \\ 0 & \frac{1}{s+5} \end{bmatrix}$$

$$e^{At} = L^{-1}\left[sI - A\right]^{-1} = \begin{bmatrix} e^{-4t} & e^{-4t} - e^{-5t} \\ 0 & e^{-5t} \end{bmatrix}$$

$$x(t) = e^{At}x(0)$$

$$= \begin{bmatrix} e^{-4t} & e^{-4t} - e^{-5t} \\ 0 & e^{-5t} \end{bmatrix} \begin{bmatrix} 5 \\ -5 \end{bmatrix}$$

$$x(t) = \begin{bmatrix} 5e^{-5t} \\ -5e^{-5t} \end{bmatrix}$$

$$X_{ss} = \lim_{t \to \infty} x(t) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Example-22.

Find the response of the system given below.

$$T(s) = \frac{-5s + 25}{s + 400}$$

$r(t) = 20u(t)$

$y(0) = 0$

Solution:

r(t) = input and y(t) = output

$$T(s) = \frac{Y(s)}{R(s)} = \frac{-5s + 25}{s + 400}$$

Dividing numerator by denominator as they are of same order-

$$\begin{array}{r|l} & -5 \\ \hline s + 400 & -5s + 25 \\ & -5s - 2000 \\ \hline & 2025 \end{array}$$

$$\frac{Y(s)}{R(s)} = -5 + \frac{2025}{s+400}$$

$$\dot{x}_1 = r(t) - 400x_1$$

and $y(t) = 2025\ x_1 - 5r(t)$

$A = [-400],\ B = [1],\ C = [2025] D = [-5]$

$$[sI - A] = [s + 400]$$

$$[sI - A]^{-1} = \left[\frac{1}{s+400}\right]$$

$$e^{At} = \left[e^{-400t}\right]$$

$y(0) = 0$, substituting in $y(t)$ -

$y(0) = 2025x_1(0) - 5r(0), \{r(0) = 20$ as given to be $20u(t)\}$

$$0 = 2025\, x_1(0) - 100$$

$$x_1(0) = \frac{100}{2025}$$

$$ZIR = e^{At}x_1(0) = \frac{100}{2025}e^{-400t}$$

$$ZSR = L^{-1}[\phi(s)BU(s)]$$

$$= L^{-1}\left[\frac{1}{s+400}.1.\frac{20}{s}\right] \quad \left\{\because\ U(s) = R(s) = \frac{20}{s}\right\}$$

$$= L^{-1}\left[\frac{20}{s(s+400)}\right] = L^{-1}\left[\frac{\frac{1}{20}}{s} - \frac{\frac{1}{20}}{s+400}\right]$$

$$ZSR = \frac{1}{20} - \frac{1}{20}e^{-400t}$$

$$x(t) = ZIR + ZSR = \frac{100}{2025}e^{-400t} + \frac{1}{20} - \frac{1}{20}e^{-400t}$$

$$x(t) = \frac{4}{81}e^{-400t} + \frac{1}{20} - \frac{1}{20}e^{-400t}$$

$$x(t) = \frac{1}{20} - 6.2 \times 10^{-4}e^{-400t}$$

$$y(t) = 2025\,x(t) - 5r(t)$$

$$y(t) = 2025\left[\frac{1}{20} - 6.2\times10^{-4}e^{-400t}\right] - 5\times20$$

$$y(t) = 1.25 - 1.25\,e^{-400t}$$

Controllability and Observability

Controllability- A system is said to be completely state controllable if it is possible to transform the system state from any initial state $x(t_0)$ to any other desired state $x(t_f)$ in a specified finite time interval $t_f - t_0$ by control vector u(t).

Note: Concept of controllability and observability were originally introduced by Kalman. Hence, Kalman's tests is used to find out whether the system is controllable and observable or not.

Kalman's Test for Controllability

Let,

$$\dot{x} = A\,x(t) + B\,u(t)$$

Then, composite matrix Q_c is given by-

$$Q_c = \left[B\ AB\ A^2B\ .....A^{n-1}B\right]$$

Condition for state controllability in s-plane

In s-plane, system is represented by transfer function. Such a system is completely state controllable only if the denominator and numerator polynominals of the transfer function do not have a common factor except the constant.

Gilbert's Test for Controllability

$$\dot{x} = A\,x(t) + B\,u(t)$$

Where, A is not in the canonical form. It can be then transformed to canonical by transformation.

$$x(t) = M\,z(t)$$

M = Model Matrix

$$\dot{z}(t) = \wedge z(t) + \tilde{B}\,u(t)$$

$\wedge$ = Diagonal Matrix

$$\tilde{B} = M^{-1}B$$

Note: The vector matrix $\tilde{B}$ should not have any zero element.

Output Controllability-

$$\dot{x}(t) = Ax(t) + Bu(t)$$

$$y(t) = Cx(t) + Du(t)$$

The system is completely output controllable if it is possible to construct an unconstrained input vector u(t) which will transfer any given initial output $y(t_0)$ to any final output $y(t_f)$ in a finite interval $0 \le t \le t_f - t_0$.

Thus, if the order of matrix C is p × n i.e. there are p outputs then for the complete output controllability, the rank of test matrix Q_c must be p.

Observability

A system is said to be completely observable, if every state $x(t_0)$ can be completely identified by measurement of output y(t) over a finite interval.

Kalman's test for observability-

$$\dot{x} = A\,x(t) + B\,u(t)$$

And $y(t) = C\,x(t)$ $\quad \left\{\begin{array}{l} \text{where, } y(t) = p \times 1 = \text{output vector} \\ \text{and } C = 1 \times n = \text{output matrix} \end{array}\right\}$

The composite matrix Q_0 is given by-

$$Q_0 = \left[C^T \;\; A^T C^T \;\; ..... \left(A^T\right)^{n-1} C^T\right]$$

C^T = Transpose of matrix C

A^T = Transpose of matrix A

Rack your Brain



Which test is used to find controllability and observability?

Condition for state observability in s-plane

There should not be cancellation of pole and zero in transfer function and numerator and denominator polynomials of transfer function must be co-prime.

Gilbert's test for observability

$$\dot{x}(t) = A\,x(t) + B\,u(t)$$

$$y(t) = C\,x(t)$$

Q by using transformation x(t) = M Z(t) where, M is model matrix.

$$\therefore y(t) = CMz(t) = \tilde{C}z(t)$$

Where,

$$\tilde{C} = CM$$

System is completely observable if all the coefficient of $\tilde{C}$ are non-zero.

Example-23.

Find the controllability of the system.

$$\dot{x} = \begin{bmatrix} 1 & 2 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$

Solution:

$$A = \begin{bmatrix} 1 & 2 \\ 1 & -1 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\therefore AB = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

$$\therefore Q_c = \begin{bmatrix} B & AB \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ 1 & -1 \end{bmatrix}$$

Now $\begin{vmatrix} 0 & 2 \\ 1 & -1 \end{vmatrix} = -2$ which is non-zero rank of $Q_c = 2 = n$

Hence, system is completely controllable.

Example-24.

Consider a system with state equation-

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -5 & -9 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u(t)$$

Estimate the state controllability by Kalman's test.

Solution:

$$Q_C = \begin{bmatrix} B & AB & A^2B \end{bmatrix} \ldots\ldots n = 3$$

$$AB = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -5 & -9 & -5 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ -5 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -5 & -9 & -5 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -5 & -9 & -5 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 0 & 0 & 1 \\ -5 & -9 & -5 \\ 25 & 40 & 16 \end{bmatrix}$$

Gray Matter Alert!!!

For a system to be completely state controllable, the necessary condition is that the rank of the composition matrix Q_c is 'n'.

If rank of $Q_c = n \rightarrow$ System is controllable

If rank of $Q_c \neq n \rightarrow$ System is not controllable

If $|Q_c| = 0 \rightarrow$ System is not controllable

If $|Q_c| \neq 0 \rightarrow$ System is controllable

$$A^2B = \begin{bmatrix} 0 & 0 & 1 \\ -5 & -9 & -5 \\ 25 & 40 & 16 \end{bmatrix}\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -5 \\ 16 \end{bmatrix}$$

$$Q_c = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & -5 \\ 1 & -5 & 16 \end{bmatrix}$$

$$\begin{vmatrix} 0 & 0 & 1 \\ 0 & 1 & -5 \\ 1 & -5 & 16 \end{vmatrix} = -1 \neq 0$$

Hence, the rank of Q_c is 3 which is 'n' i.e. system is completely controllable.

Example-25.

Consider the system with state equation-

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u(t)$$

Estimate the state controllability using Gilbert's test.

Solution:

Firstly convert A in to canonical form.

$$|\lambda I - A| = \begin{bmatrix} \lambda & -1 & 0 \\ 0 & \lambda & -1 \\ 6 & 11 & \lambda + 6 \end{bmatrix} = 0$$

$$\therefore \lambda^3 + 6\lambda^2 + 11\lambda + 6 = 0$$

$$\therefore (\lambda + 1)(\lambda + 2)(\lambda + 3) = 0$$

$$\lambda_1 = -1, \lambda_2 = -2, \lambda_3 = -3$$

As matrix A is in phase variable form, the model matrix M is-

$$M = \begin{bmatrix} 1 & 1 & 1 \\ \lambda_1 & \lambda_2 & \lambda_3 \\ \lambda_1^2 & \lambda_2^2 & \lambda_3^2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ -1 & -2 & -3 \\ 1 & 4 & 9 \end{bmatrix}$$

Gray Matter Alert!!!

Rank of $Q_0 = n \rightarrow$ System is completely observable

Rank of $Q_0 \neq n \rightarrow$ system is not-completely observable

$|Q_0| = 0 \rightarrow$ system is not observable

$|Q_0| \neq 0 \rightarrow$ system is observable

$$M^{-1} = \frac{\text{Adj}\,|M|}{|M|} = \frac{\begin{bmatrix} -6 & 6 & -2 \\ -5 & 8 & -3 \\ -1 & 2 & -1 \end{bmatrix}^T}{-2} = \frac{\begin{bmatrix} -6 & -5 & -1 \\ 6 & 8 & 2 \\ -2 & -3 & -1 \end{bmatrix}}{-2}$$

$$= \begin{bmatrix} 3 & 2.5 & 0.5 \\ -3 & -4 & -1 \\ 1 & 1.5 & 0.5 \end{bmatrix}$$

$$\tilde{B} = M^{-1}B = \begin{bmatrix} 3 & 2.5 & 0.5 \\ -3 & -4 & -1 \\ 1 & 1.5 & 0.5 \end{bmatrix}\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.5 \\ -1 \\ 0.5 \end{bmatrix}$$

None of the elements of $\tilde{B}$ is zero. So, it is completely controllable.

Note:- Gilbert's test is rarely used because it is time consuming.

Example-26.

Evaluate the observability of the system-

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -3 & -4 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \text{ and } C = \begin{bmatrix} 4 & 5 & 1 \end{bmatrix}$$

Solution:

Order of the system is n = 3.

$$\therefore Q_0 = \begin{bmatrix} C^T & A^TC^T & \left(A^T\right)^2 C^T \end{bmatrix}$$

$$A^T = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & -3 \\ 0 & 1 & -4 \end{bmatrix}$$

$$C^T = \begin{bmatrix} 4 \\ 5 \\ 1 \end{bmatrix};$$

$$A^TC^T = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & -3 \\ 0 & 1 & -4 \end{bmatrix}\begin{bmatrix} 4 \\ 5 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

Previous Year's Questions

The second order dynamic system $\frac{dX}{dt} = PX + QU,\ Y = RX$ has the matrices P, Q and R as follows:

$$P = \begin{bmatrix} -1 & 1 \\ 0 & -3 \end{bmatrix}, Q = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \text{ and } R = \begin{bmatrix} 0 & 1 \end{bmatrix}.$$

The system has the following controllability and observability properties:

(a) Controllable and observable
(b) Non controllable but observable
(c) Controllable but not observable
(d) Non controllable and not observable

Ans (c)

$$\left(A^T\right)^2 C^T = A^T\left[A^T C^T\right] = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & -3 \\ 0 & 1 & -4 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -3 \\ -3 \end{bmatrix}$$

$$Q_0 = \begin{bmatrix} 4 & 0 & 0 \\ 5 & 1 & -3 \\ 1 & 1 & -3 \end{bmatrix}$$

Consider the determinant,

$$\begin{vmatrix} 4 & 0 & 0 \\ 5 & 1 & -3 \\ 1 & 1 & -3 \end{vmatrix} = 0$$

$\therefore$ Rank of $Q_0 \neq 3 \neq n$

Hence, the system is not completely observable.

Gray Matter Alert!!!

Short trick: If system matrix [A] is a diagonal matrix then-
(i) System is observable, if the output matrix [C] contains any non-zero element.
(ii) The system is controllable, if the input matrix [B] contains any non-zero element.

Example-27.
Evaluate the observability of the system using Gilbert's test.

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -3 & -4 \end{bmatrix},\ B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \text{ and } C = \begin{bmatrix} 4 & 5 & 1 \end{bmatrix}$$

Solution:

$$\left[\lambda I - A\right] = \begin{vmatrix} \lambda & -1 & 0 \\ 0 & \lambda & -1 \\ 0 & 3 & \lambda + 4 \end{vmatrix} = 0$$

$$\therefore \lambda\left(\lambda\left(\lambda + 4\right) + 3\right) + 0 + 0 = 0$$

$$\lambda^3 + 4\lambda^2 + 3\lambda = 0$$

$$\lambda\left(\lambda^2 + 4\lambda + 3\right) = 0$$

$$\lambda\left(\lambda + 1\right)\left(\lambda + 3\right) = 0$$

$$\lambda_1 = 0,\ \lambda_2 = -1,\ \lambda_3 = -3$$

Eigen values are distinct so, model matrix M is-

$$M = \begin{vmatrix} 1 & 1 & 1 \\ \lambda_1 & \lambda_2 & \lambda_3 \\ \lambda_1^2 & \lambda_2^2 & \lambda_3^2 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ 0 & -1 & -3 \\ 0 & 1 & 9 \end{vmatrix}$$

$$\tilde{C} = CM = \begin{bmatrix} 4 & 5 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & -3 \\ 0 & 1 & 9 \end{bmatrix}$$

$$\tilde{C} = \begin{bmatrix} 4 & 0 & -2 \end{bmatrix}$$

As there is one zero element in $\tilde{C}$, the system is not completely observable.

Example-28.

Check controllability and observability.

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ -40 & -18 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

$$y = [1 \quad 0]x$$

Solution:

Controllability-

$$B = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \qquad AB = \begin{bmatrix} 0 & 1 \\ -40 & -18 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -18 \end{bmatrix}$$

$$Q_C = \begin{bmatrix} B & AB \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & -18 \end{bmatrix}, \ |Q_c| \neq 0$$

$\therefore$ System is controllable.

Observability-

$$A^T C^T = \begin{bmatrix} 0 & -40 \\ 1 & -18 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$Q_0 = [C^T \quad A^T C^T] = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, |Q_0| \neq 0$$

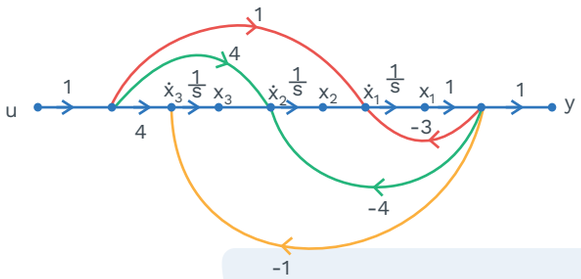
$\therefore$ System is observable.

Example-29.

Construct signal flow graph and state model for system whose transfer function is $T(s) = \dfrac{s^2 + 4s + 4}{s^3 + 3s^2 + 4s + 1}$

Solution:

$$T(s) = \frac{\frac{1}{s} + \frac{4}{s^2} + \frac{4}{s^3}}{1 - \left(-\frac{3}{s} - \frac{4}{s^2} - \frac{1}{s^3}\right)}$$



$y = x_1$

$\dot{x}_1 = x_2 - 3y + u = x_2 - 3x_1 + u = -3x_1 + x_2 + u$

$\dot{x}_2 = x_3 - 4y + 4u = x_3 - 3x_1 + 3u = -4x_1 + x_3 + 4u$

$\dot{x}_3 = 4u - y = 4u - x_1 = -x_1 + 4u$

So state model is-

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -3 & 1 & 0 \\ -4 & 0 & 1 \\ -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 \\ 4 \\ 4 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

PRACTICE QUESTIONS

Q1 For a system represented by $\dot{x} = Ax$, the response is $x(t) = \begin{bmatrix} 2e^{-4t} \\ e^{-4t} \end{bmatrix}$ when $x(0) = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $\dot{x}(t) = \begin{bmatrix} 4e^{-2t} \\ e^{-2t} \end{bmatrix}$ when $x(0) = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$

The system matrix A will be-

(a) $\begin{bmatrix} 0 & -8 \\ 1 & -6 \end{bmatrix}$ (b) $\begin{bmatrix} -8 & 0 \\ 1 & -6 \end{bmatrix}$ (c) $\begin{bmatrix} 0 & -8 \\ -6 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 0 & 8 \\ -1 & 6 \end{bmatrix}$

Sol: (a)

Q2 The state space representation of system is given by-

$\dot{x} = \begin{bmatrix} 0 & 1 \\ 0 & -3 \end{bmatrix} x + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u, y = \begin{bmatrix} 1 & 0 \end{bmatrix} x$. The transfer function $\frac{Y(s)}{U(s)}$ of the system will be-

(a) $\frac{1}{s+3}$ (b) $\frac{1}{s}$ (c) $\frac{1}{s(s+3)}$ (d) $\frac{1}{s^2}$

Sol: (b)

Q3 A network is described by the state model as-

$\frac{dx_1}{dt} = 2x_1 - x_2 + 3u, \frac{dx_2}{dt} = -4x_2 - u, y = 3x_1 - 2x_2$

The transfer function $H(s) = \frac{Y(s)}{U(s)}$ is-

(a) $\frac{11s + 35}{(s-2)(s+4)}$ (b) $\frac{11s - 35}{(s-2)(s+4)}$

(c) $\frac{11s + 38}{(s-2)(s+4)}$ (d) $\frac{11s - 35}{(s+2)(s-4)}$

Sol: (a)

Q4 A second order LTI system is described by the following state equations-

$$\frac{d}{dt}x_1(t) - x_2(t) = 0$$

$$\frac{d}{dt}x_2(t) + 2x_1(t) + 3x_2(t) = r(t)$$

When $x_1(t)$ and $x_2(t)$ are the two state variables and $r(t)$ denotes the input. The output $c(t) = x_1(t)$, then the system is-

(a) Undamped
(b) Under damped
(c) Critically Damped
(d) Over damped

Sol: (d)

Q5 Match list-I: matrix with list-II: dimension for the state equation.

$$\dot{x}(t) = Px(t) + Qu(t)$$

$$y(t) = Rx(t) + Su(t)$$

List-I (Matrix)	List-II (Dimension)
(A) P	(1) $n \times p$
(B) Q	(2) $q \times n$
(C) R	(3) $q \times p$
(D) S	(4) $p \times p$
	(5) $n \times n$

	A	B	C	D
(a)	1	2	3	4
(b)	5	1	4	2
(c)	5	1	2	3
(d)	4	1	2	3

Sol: (c)

Q6 **Find the eigen values for the matrix-**

$$[A]=\begin{bmatrix}0 & 0\\ -2 & -4\end{bmatrix}$$

(a) $\lambda_1=0,\lambda_2=-4$

(b) $\lambda_1=2,\lambda_2=2$

(c) $\lambda_1=2,\lambda_2=-2$

(d) $\lambda_1=0,\lambda_2=0$

Sol: (a)

Q7 **Given state model of a system is-**

$$\dot{x}=\begin{bmatrix}0 & 1\\ -4 & -5\end{bmatrix}x+\begin{bmatrix}0\\ 4\end{bmatrix}u$$

$$y=\begin{bmatrix}1 & 0\end{bmatrix}x$$

with initial condition $x(0)=\begin{bmatrix}1\\ 1\end{bmatrix}$

The state transition matrix is-

(a) $\begin{bmatrix}\frac{4}{3}e^{-t}-\frac{1}{3}e^{-4t} & \frac{1}{3}e^{-t}-\frac{1}{3}e^{-4t}\\ \frac{-4}{3}e^{-t}+\frac{4}{3}e^{-4t} & \frac{-1}{3}e^{-t}+\frac{4}{3}e^{-4t}\end{bmatrix}$

(b) $\begin{bmatrix}\frac{4}{3}e^{-t}+\frac{1}{3}e^{-4t} & \frac{1}{3}e^{-t}+\frac{1}{3}e^{-4t}\\ \frac{-4}{3}e^{-t}+\frac{4}{3}e^{-4t} & \frac{-1}{3}e^{-t}+\frac{4}{3}e^{-4t}\end{bmatrix}$

(c) $\begin{bmatrix}\frac{4}{3}e^{-t}+\frac{1}{3}e^{-4t} & \frac{1}{3}e^{-t}+\frac{1}{3}e^{-4t}\\ \frac{4}{3}e^{-t}-\frac{4}{3}e^{-4t} & \frac{1}{3}e^{-t}-\frac{4}{3}e^{-4t}\end{bmatrix}$

(d) 0

Sol: (a)

Q8 A system is described by the following equation,

$$\dot{x}(t)=\begin{bmatrix}0 & 0 & 0\\ 1 & 0 & -3\\ 0 & 1 & -4\end{bmatrix}x(t)+\begin{bmatrix}40\\ 10\\ 0\end{bmatrix}u(t)$$

$y(t) = [1 \quad 0]x(t)$

whose controllability matrix is $\begin{bmatrix}40 & 0 & a\\ 10 & b & c\\ 0 & 10 & 0\end{bmatrix}$. The sum of a, b, c is-

(a) 10 (b) -10 (c) 70 (d) -70

Sol: (a)

Q9 For the given closed loop transfer function $\frac{C(s)}{R(s)}=\frac{2(s+3)}{(S+1)(s+2)}$,

the state model is-

(a) $\begin{bmatrix}\dot{x}\end{bmatrix}=\begin{bmatrix}-1 & 0\\ -2 & -1\end{bmatrix}x(t)+\begin{bmatrix}1\\ 1\end{bmatrix}u(t)$

$y(t)=\begin{bmatrix}-4 & 2\end{bmatrix}x(t)$

(b) $\begin{bmatrix}\dot{x}\end{bmatrix}=\begin{bmatrix}-1 & 0\\ 0 & -2\end{bmatrix}x(t)+\begin{bmatrix}1\\ 1\end{bmatrix}u(t)$

$y(t)=\begin{bmatrix}+4 & -2\end{bmatrix}x(t)$

(c) $\begin{bmatrix}\dot{x}\end{bmatrix}=\begin{bmatrix}-2 & 0\\ 0 & -1\end{bmatrix}x(t)+\begin{bmatrix}1\\ 0\end{bmatrix}u(t)$

$y=\begin{bmatrix}4 & -2\end{bmatrix}x(t)$

(d) $\begin{bmatrix}\dot{x}\end{bmatrix}=\begin{bmatrix}-2 & 0\\ 0 & -1\end{bmatrix}x(t)+\begin{bmatrix}1\\ 1\end{bmatrix}u(t)$

$y=\begin{bmatrix}-4 & -2\end{bmatrix}x(t)$

Sol: (b)

Q10 For the given differential equation-

$$\dddot{y} + 6\ddot{y} + 5\dot{y} + y = u$$

The matrix [A] is-

(a) $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 5 & 6 \end{bmatrix}$

(b) $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -5 & -1 \end{bmatrix}$

(c) $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -5 & -6 \end{bmatrix}$

(d) $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 5 & 6 \end{bmatrix}$

Sol: (c)

Q11 For the given equation $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -6 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$ and $y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ with input being unit step and initial conditions be 0. The zero state response is-

(a) $\frac{1}{6} - \frac{1}{2}e^{-2t}$

(b) $\frac{1}{6} - \frac{1}{2}e^{-2t} + \frac{1}{3}e^{-3t}$

(c) 0

(d) $\frac{1}{6} + \frac{1}{3}e^{-3t}$

Sol: (b)